Quantum Technology for Comparator Circuit

Hafiz Md. Hasan Babu[1], Khandaker Mohammad Mohi Uddin[2], Rownak Borhan Himel[2], and Nitish Biswas[2]
[1]Dept. of Computer Science and Engineering, Dhaka University, Dhaka, Bangladesh
[2]Dept. of Computer Science and Engineering, Dhaka International University, Dhaka, Bangladesh
[1]E-mail: hafizbabu@du.ac.bd

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Abstract

Quantum computing is the fusion of quantum physics and theoretical computer science that makes it more special and super faster than other traditional silicon-based computing systems. In this work, we have designed quantum 2-bit string comparator using the basic quantum AND, OR, NOT, XOR operations that are based on basic quantum CNOT, V, V+ gate operations. The proposed quantum comparator take the advantages of basic quantum computing to perform parallel processing and super-fast calculations.

Keywords

Quantum computing, silicon-based, string comparators

1. Introduction

A number of intractable classical computers problems are solved by quantum computing and that makes it an interesting concept for computing techniques [1]. Quantum circuits are used to design quantum computing-based systems [2] and while designing quantum circuits, quantum gates are required. A lot of algorithms are used in quantum computation these are faster than their traditional counterparts [3]. In this work, 2-bit string comparator is designed using basic quantum AND, OR, NOT operations.

2. Proposed quantum comparator

A magnitude digital comparator takes two binary or digital numbers and generates three outputs whether one number is less than, equal or greater than another number.

Let, for 2-bit string comparator, Inputs are: $A = A1A2$ and $B = B1B2$; Outputs are X, Y, Z.

$$X = A > B \quad (1)$$

$$Y = A\ equal\ B \quad (2)$$

$$Z = A < B \quad (3)$$

X, Y and Z can be obtained utilizing the following equations:

$$X = B2'.A2 + A1.A2.B1' + A1.B1'.B2' \quad (4)$$

$$Y = (A2\ XNOR\ B2).(A1\ XNOR\ B1) \quad (5)$$

$$Z = A1.A2' + B1.A2'.A1' + A1'.B2.B1 \quad (6)$$

2-Bit string quantum comparator is designed using 4 qubits. Here, two qubits represent a string and another two qubits represent another string to compare with the first one. We have designed basic quantum AND, OR operations suing basic quantum CNOT, V and V+ gates. “Fig. 1” shows the proposed 2-bit string quantum comparator. “Equation 4” to “Equation 6” help to create the proposed circuit. Here, |A1> |A2> is the first bit string and |B1>|B2> is the second bit string. The value of |A1>, |A2>, |B1>, |B2> can be |1> or |0>.

Suppose, for 2-input strings like |0>|1> and |0>|0>, |Y> and |Z> will be |0> and |X> will be |1>. Because, |0>|1> is greater than |0>|0>. And for the input strings |0>|0> and |0>|0> or |1>|1> and |1>|1>, |X> and |Z> will be |0> and |Y> will be |1>. Because, when both inputs are equal and |Y> will be |1>.

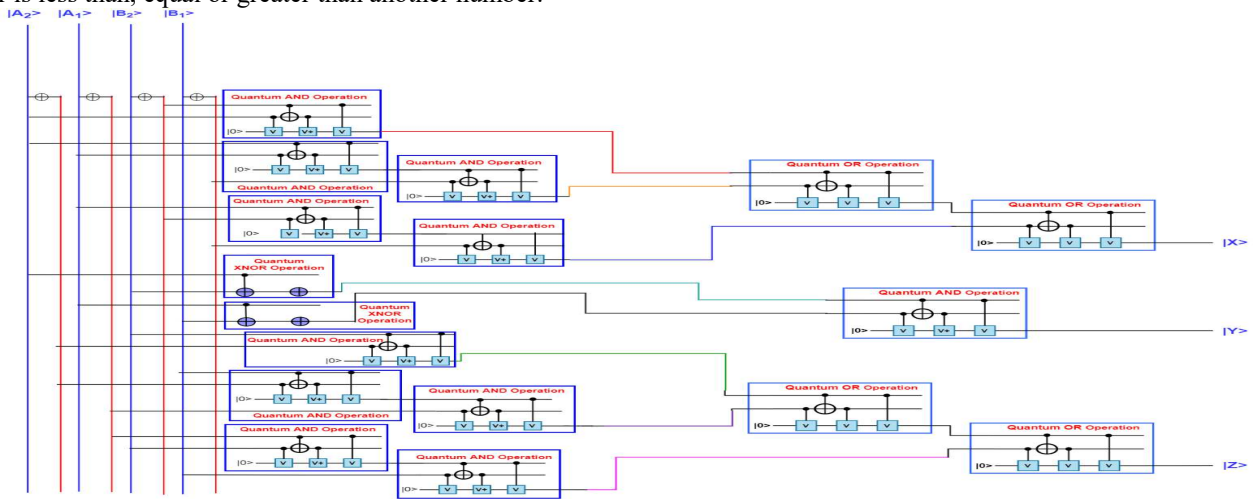


Fig. 1: Proposed quantum comparator

3. Conclusions

In this work, 2-bit string quantum comparator is designed using basic quantum CNOT, V, V+ gate operations. The proposed comparator circuit is quite faster than a traditional comparator circuit that can be used in artificial intelligence, financial modeling, and cybersecurity.

References

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